

$$\textcircled{1} \textcircled{a} \max_{c, l} \ln(c) + \frac{b}{1-\delta} l^{1-\delta}$$

s.t:

$$c \leq w(1-\tau) \cdot \underbrace{(1-l)}_{\substack{\text{employed} \\ \text{time} \rightarrow n}} + T$$

F.O.C:

$$c \cdot l^{-\delta} = \frac{1}{b} \cdot w(1-\tau) \quad \# \text{Consumption-leisure tradeoff}$$

$$\text{Labor: } n = 1-l \rightarrow c \cdot (1-n)^{-\delta} = \frac{1}{b} \cdot w(1-\tau)$$

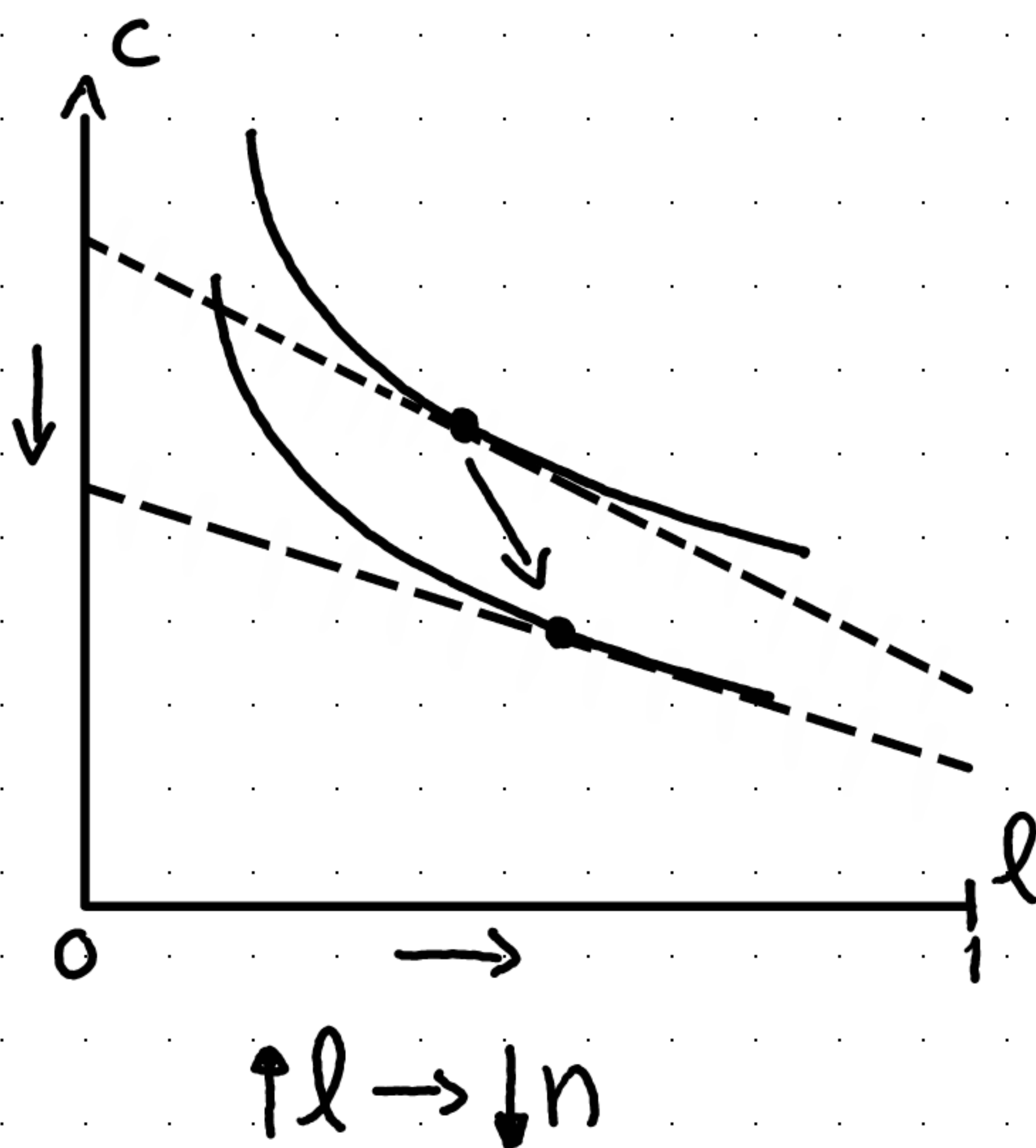
$$\underbrace{(w(1-\tau) \cdot n + T)}_c \cdot (1-n)^{-\delta} = \frac{w}{b} (1-\tau)$$

$$(1-n)^{\delta} - b \cdot n = \frac{b \cdot T}{(1-\tau)w}$$

$\textcircled{b}$

$$\underbrace{(1-n)^{\delta} - b \cdot n}_{n \downarrow} = \frac{b \cdot T}{(1-\tau)w} \quad \uparrow$$

Income and substitution
eff - + - - - $l \leftrightarrow n$ connection



② a)

$$E_{t+1} = \underbrace{E_t - s \cdot E_t}_{(1-s)E_t} + \underbrace{m(V_t, U_t)}_{\mu V_t^\alpha U_t^{1-\alpha}}$$

$\mu \rightarrow$ Match efficiency $\rightarrow$ Increases inflow of workers from unemployment to employment

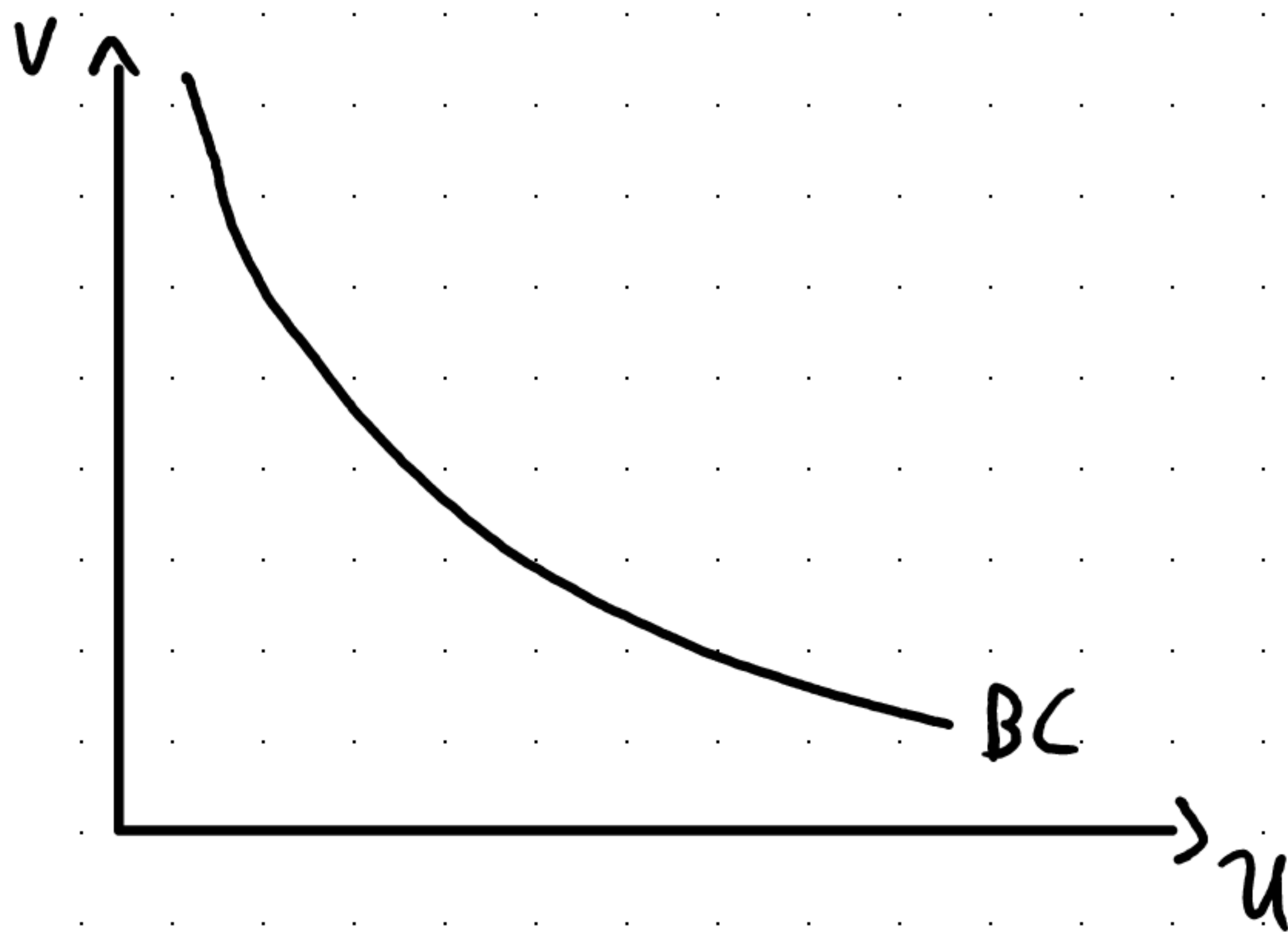
b)

$$U_t = L_t - E_t \rightarrow U_{t+1} = U_t + s \cdot E_t - m \cdot (V_t, U_t) \rightarrow \text{similar to } E_{t+1}$$

$$\text{In SS: } \frac{U}{L} = \frac{U}{L} + \frac{s \cdot E}{L} - \frac{\mu V^\alpha U^{1-\alpha}}{L}$$

$$\therefore u = u + s \cdot (1-u) - \mu v^\alpha \cdot u^{1-\alpha}$$

$$v(u) = \left[\frac{s(1-u)}{\mu \cdot u^{1-\alpha}} \right]^{\frac{1}{\alpha}}$$



BC $\rightarrow$ job creation = job destruction

$$c) \quad \frac{m(V, U)}{V} = \frac{M \cdot V^\alpha U^{1-\alpha}}{V} = M \cdot \left(\frac{U}{V}\right)^{1-\alpha} = \boxed{M \left(\frac{1}{\Theta}\right)^{1-\alpha} = q}$$

$\hookrightarrow \frac{1}{\Theta}$

$$\frac{m(V, U)}{U} = \frac{M V^\alpha U^{1-\alpha}}{U} = M \cdot \left(\frac{V}{U}\right)^\alpha = \boxed{M \cdot \Theta^\alpha = f}$$

$\hookrightarrow \Theta$

$$\left. \begin{array}{l} q = M \cdot \Theta^{\alpha-1} \\ f = M \cdot \Theta^\alpha \end{array} \right\} M \cdot \Theta^\alpha \cdot \frac{\Theta}{\Theta} = \underbrace{M \cdot \Theta^{\alpha-1}}_q \cdot \Theta \rightarrow \boxed{f = q \cdot \Theta}$$

$$\frac{df}{d\Theta} = M \cdot \alpha \cdot \Theta^{\alpha-1} > 0$$

$$\frac{dq}{d\Theta} = M(\alpha-1) \cdot \left(\frac{1}{\Theta^\alpha}\right) < 0$$

$\underbrace{\alpha-1}_{<0}$

$\uparrow \Theta \rightarrow$ easier job finding for workers (f), but vacancies are harder to fill $\rightarrow$ more competition + congestion

d

$$f = \mu \cdot \Theta^\alpha = \mu \left(\frac{V}{U}\right)^\alpha \rightarrow \frac{df}{dV} = \frac{\mu \cdot \alpha \cdot \left(\frac{V}{U}\right)^{\alpha-1}}{V} > 0$$

$$q = \mu \cdot \Theta^{\alpha-1} = \mu \left(\frac{V}{U}\right)^{\alpha-1} \rightarrow \frac{dq}{dV} = \frac{\mu \cdot (\alpha-1) \cdot \Theta^{\alpha-2}}{V} < 0$$

Looking at total matches:

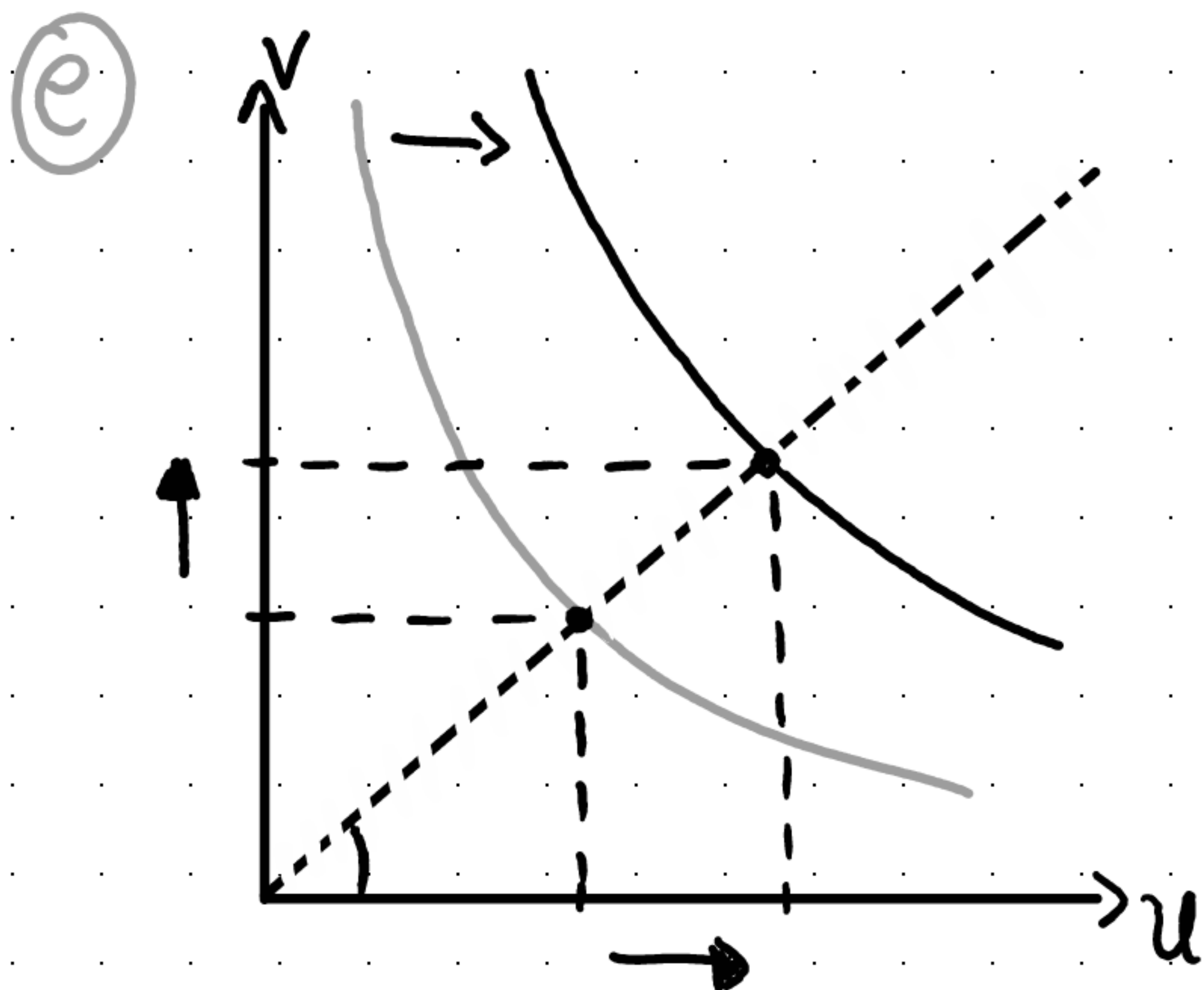
$$m = \mu \cdot V^\alpha U^{1-\alpha} \rightarrow \frac{\partial m}{\partial V} = \mu \cdot \alpha \left(\frac{V}{U}\right)^{\alpha-1} = \mu \cdot \alpha \cdot \Theta^{\alpha-1} = \boxed{\alpha \cdot q}$$

Matches increase less, depend on α and the job filling rate

↳ Vacancies to employment is not 1:1 → friction

Congestion externality → Firm Posting the Vacancy is better off, but others are "harmd"

↳ Workers also benefit from higher chance to find a job



Less efficient matching $\rightarrow$ More v per u in SS

Shift in all v - u values and possible states rather than just changing how Θ is split between u and v